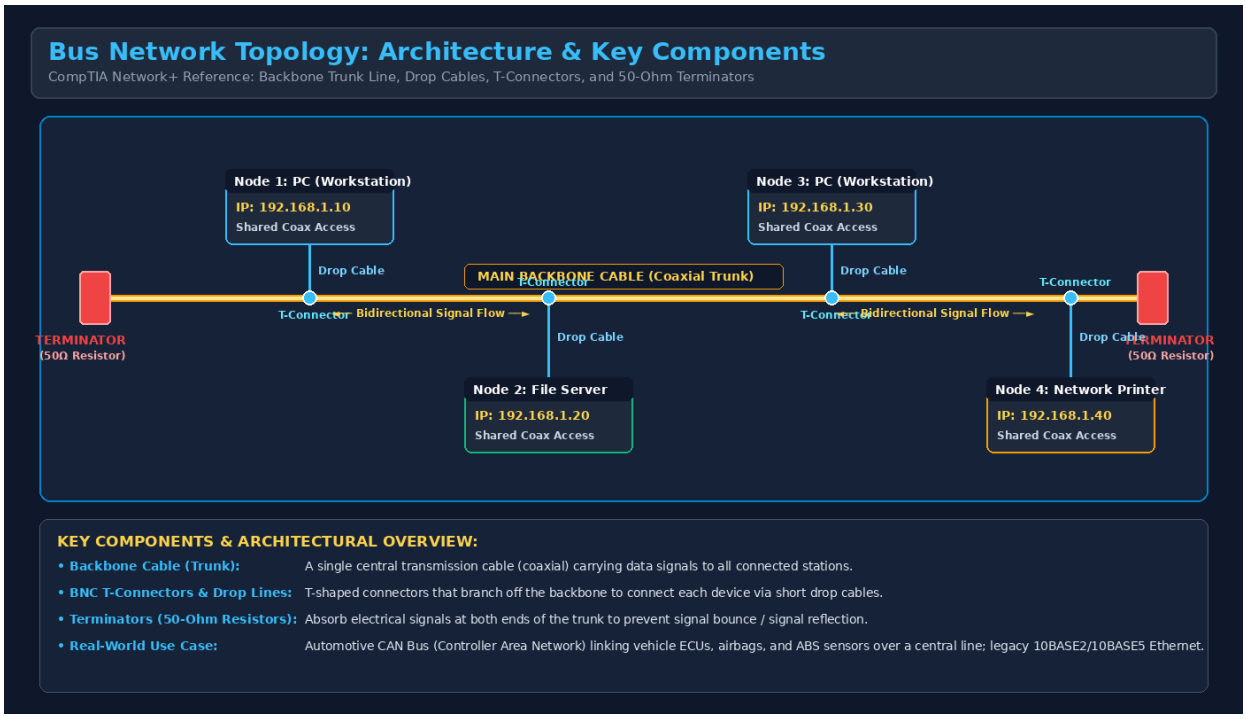
Network Topologies Theory

Task 1: Bus Topology Diagram, Key Components & Real-World Example

Key Components:



1. **Backbone Cable (Trunk line):** A single main transmission cable (traditionally thick or thin coaxial cable like RG-58) that carries data signals across the entire network.
2. **Drop Lines & T-Connectors (BNC):** T-connectors tap into the main cable to link individual workstations (nodes) to the trunk line via short drop cables.
3. **Terminators (50-Ohm Resistors):** Placed at both physical ends of the backbone cable to absorb electrical signals, preventing **signal bounce / reflection** which corrupts data packets.
4. **Nodes:** End devices (computers, file servers, network printers) sharing the single communication channel.

Real-World Example:

- **Automotive CAN Bus (Controller Area Network):** Modern cars use a high-reliability serial bus topology to interconnect microcontrollers, sensors, braking systems (ABS), and the engine control unit (ECU) along a single two-wire backbone. It was also widely used in legacy **10BASE2 ("Thinnet")** Ethernet office networks.

Task 2: Cisco Packet Tracer Star Topology Setup (Simulation Steps & Screenshot)

Step-by-Step Packet Tracer Instructions:

1. **Place the Central Switch:**

Under **Network Devices $\rightarrow$ Switches**, drag and drop a **Cisco Catalyst 2960-24TT Switch** onto the workspace.

2. **Place 4 End Devices:**

Under **End Devices $\rightarrow$ PC**, place 4 desktop PCs around the switch.

3. **Rename the PCs after popular apps:**

Click on each PC's display label below its icon and rename them:

   - PC 0 $\rightarrow$ **PC-WhatsApp**
   - PC 1 $\rightarrow$ **PC-Zomato**
   - PC 2 $\rightarrow$ **PC-Spotify**
   - PC 3 $\rightarrow$ **PC-Flipkart**

4. **Connect Devices with Copper Straight-Through Cables:**

Select the **Connections (lightning bolt)** icon, choose **Copper Straight-Through cable (solid black line)**, and connect:

   - PC-WhatsApp (FastEthernet0) $\rightarrow$ Switch (FastEthernet0/1)
   - PC-Zomato (FastEthernet0) $\rightarrow$ Switch (FastEthernet0/2)
   - PC-Spotify (FastEthernet0) $\rightarrow$ Switch (FastEthernet0/3)
   - PC-Flipkart (FastEthernet0) $\rightarrow$ Switch (FastEthernet0/4)

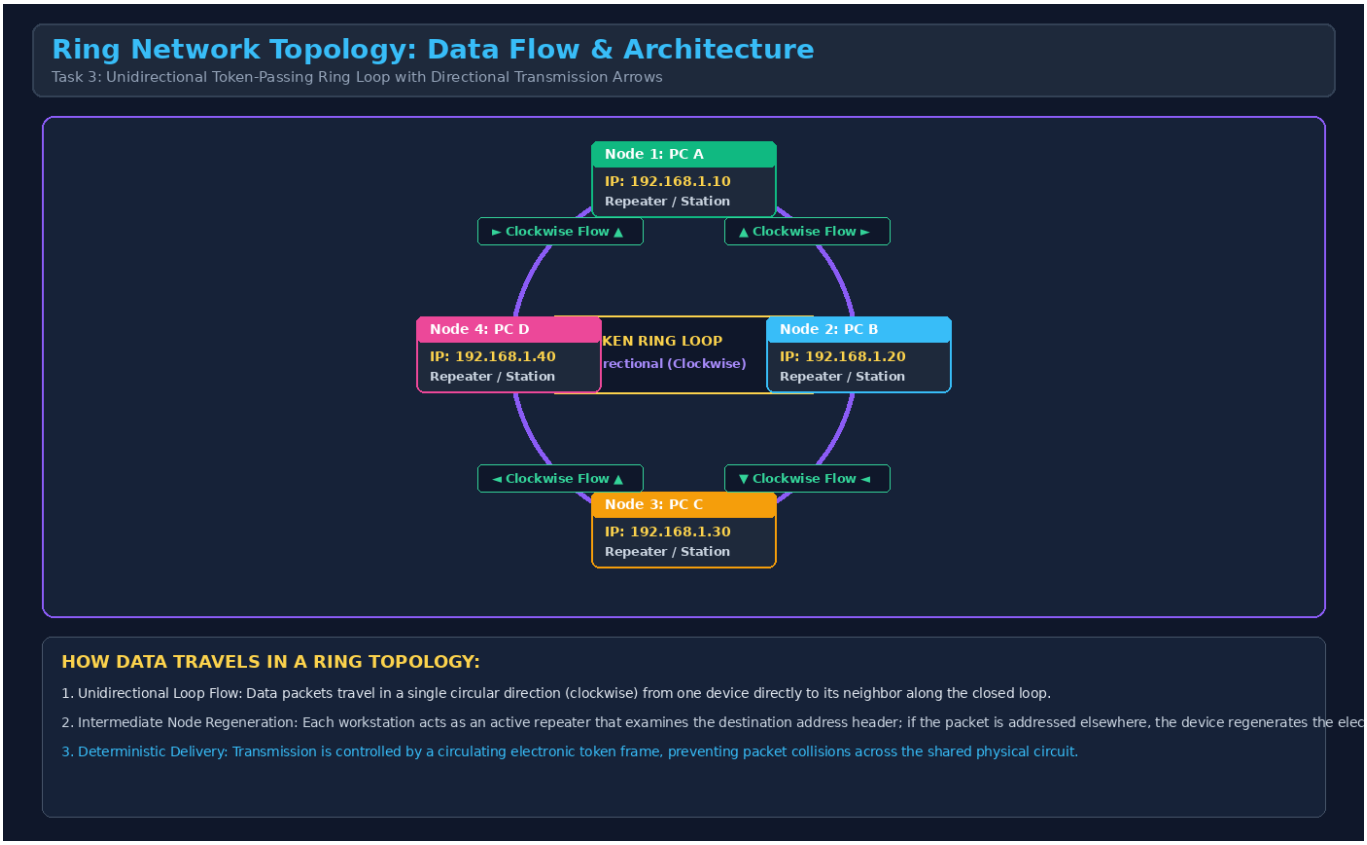
5. **Assign Static IP Addresses:**

Click each PC $\rightarrow$ **Desktop** tab $\rightarrow$ **IP Configuration**:

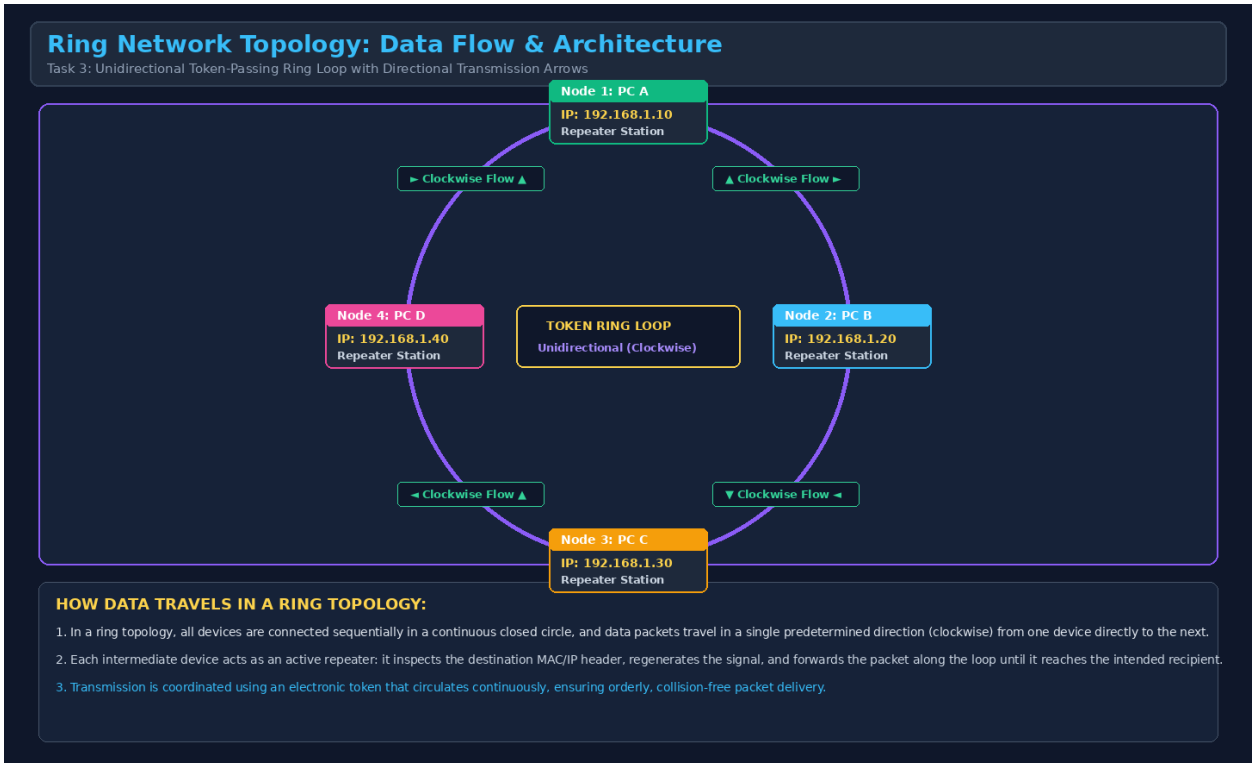
   - PC-WhatsApp: 192.168.1.10 (Subnet: 255.255.255.0)
   - PC-Zomato: 192.168.1.20 (Subnet: 255.255.255.0)
   - PC-Spotify: 192.168.1.30 (Subnet: 255.255.255.0)
   - PC-Flipkart: 192.168.1.40 (Subnet: 255.255.255.0)

6. **Verify:** Both link lights will turn from amber to solid **green**, indicating the star topology is active and ready for ping tests.

Task 3: Ring Topology Diagram & Data Transmission Flow



Task 4: Comparison Table: Mesh Topology vs. Star Topology



1. **Zero Single Point of Failure:** Every node has dedicated redundant point-to-point links; if one cable breaks, traffic automatically re-routes with zero downtime.

2. **High Security & Privacy:** Dedicated lines ensure messages travel directly between sender and receiver without intermediate node inspection. | 1. **Extremely Expensive & Complex:** Requires a massive amount of physical cabling ($N(N-1)/2$ cables for full mesh) and numerous network interface ports.

2. **Difficult Installation & Scalability:** Adding or relocating nodes requires significant re-cabling and complex administration. | | **Star Topology**

(Hub / Switch Centered) | 1. **Easy Fault Isolation & Maintenance:** If an individual cable or PC fails, only that single node goes down while the rest of the network remains fully operational.

2. **Simple Setup & Highly Scalable:** Adding new devices requires merely plugging a cable into an open switch port without interrupting active connections. | 1. **Central Point of Failure:** If the central switch or hub crashes or loses power, the entire network goes down immediately.

2. **Cable Dependency:** Requires more cabling than a basic bus topology because every single node must run a dedicated line back to the central switch. |

Complete Topologies Reference Visual Card

The diagram below combines the **Bus Topology**, **Star Topology** (matching the Packet Tracer assignment), and **Ring Topology**: